

Fabius Monomial Antiderivatives and Inverse-Quantile Integrals

Exact primitive ladders, complex-exponent series, Rvachev transforms, Mellin interpolation, and nonlinear inverse reductions

Frontier research report prepared from the recursive TeX corpus
Vladimir Reshetnikov's ProveIt repository

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Snapshot-relative novelty audit; no claim of unconditional worldwide priority

Abstract

Let F be the bounded Fabius distribution function, let $\text{up}(x) = F(1 - |x|)$ be Rvachev's compactly supported even up-function, let $\mathcal{F}$ be the signed global extension satisfying $\mathcal{F}'(x) = 2\mathcal{F}(2x)$, and let $J = F^{-1}$ on $[0, 1]$. A recursive audit of the LaTeX documents under `Analysis/FabiusFunction/docs` finds extensive treatments of exact dyadic arithmetic, moments, q -binomial and Thue–Morse formulae, repeated-integration limits, Fourier sinc products, endpoint asymptotics, and inverse asymptotics, but no systematic calculus of indefinite monomial-weighted primitives.

The central observation is the exact zero-based primitive ladder

$$A_n(x) = 2^{\binom{n}{2}} F(x/2^n), \quad A'_n = A_{n-1}, \quad A_0 = F,$$

valid for $x \leq 1$. It immediately gives, for every $p \in \mathbb{N}_0$,

$$\int x^p F(x) dx = \sum_{k=0}^p (-1)^k \frac{p!}{(p-k)!} 2^{\binom{k+1}{2}} x^{p-k} F(x/2^{k+1}) + C.$$

For arbitrary $\alpha \in \mathbb{C}$ the terminating sum becomes an absolutely and locally uniformly convergent series. Endpoint flatness is strong enough that this series is exact for every complex exponent, including all negative powers, rather than merely asymptotic. Consequently $\mathcal{M}_F(s) = \int_0^1 x^{s-1} F(x) dx$ is entire and has an explicit dyadic Newton-type expansion. Differentiation in s produces logarithmic weights, with coefficients expressible through signed Stirling numbers or complete Bell polynomials.

Reflection gives primitives of $F(1 - x) = 1 - F(x)$, and the same formulas provide piecewise primitives and all polynomial moments of up . For the inverse function a change of variables yields the exact nonlinear reduction

$$\int_0^y t^\beta J(t) dt = \frac{y^{\beta+1} J(y)}{\beta+1} - \frac{1}{\beta+1} \int_0^{J(y)} F(x)^{\beta+1} dx,$$

with a separate logarithmic identity at $\beta = -1$. Positive integer powers of F in the remainder admit an explicit finite Thue–Morse spline limit. The repository's Lambert– W inverse asymptotic then gives sharp endpoint laws for these inverse primitives, including a phase transition at $\beta = -1$. Exact rational checks and independent quasi-Monte Carlo experiments accompany the proofs. Several conjectural extensions and a concrete Lean formalization roadmap conclude the report.

Current Lean formalization status. The complete natural-order core is now machine-checked, and in greater generality than it is first presented in this report. The module `FabiusFunction.NormalizedVolterra` treats arbitrary real normed-space-valued inputs, arbitrary base points, oriented endpoints, and order zero. For continuous Banach-valued inputs, `iteratedPrimitive_eq_normalizedVolterra` identifies literal iteration with the Cauchy kernel; without a completeness assumption, `normalizedVolterra_polynomial` and `normalizedVolterra_monomial` prove the finite polynomial and natural-monomial commutators. Also for arbitrary real normed targets, the forward form `normalizedVolterra_affine` covers every real scale c , including $c = 0$ and negative c , while its inverse form `normalizedVolterra_comp_affine` assumes $c \neq 0$. `normalizedVolterra_basepoint_shift` needs interval integrability only from a to b and from b to x , has no endpoint-order hypothesis, and includes $n = 0$. `normalizedVolterra_succ_eq_taylor_of_eq_zero` retains only integrability from a to b , is positive-order, and assumes vanishing on the open unoriented interval between b and x , with neither endpoint order nor endpoint values. The module `FabiusFunction.FabiusAntiderivatives` proves the signed global ladder and monomial formula in `normalizedVolterra_extendedFabius` and `normalizedVolterra_pow_mul_extendedFabius`. Its bounded counterparts `normalizedVolterra_fabiusReal_of_le_one` and `normalizedVolterra_pow_mul_fabiusReal_of_le_one` hold on the proved initial half-line $x \leq 1$, not merely on $[0, 1]$.

These four new declarations are generic transport and finite Taylor-jet identities. They do not themselves compute explicit Fabius/up coefficients or assemble piecewise formulas. They were focused-build verified at compiled checkpoint e109088ed.

No Lean formalization is claimed here for the complex-power series or its convergence and remainder estimates, the logarithmic and Mellin consequences, fractional primitive order, the Rvachev piecewise or fractional primitives, the inverse-function formulas and asymptotics, or the bounded function's piecewise continuation for $x > 1$. Those portions remain research frontiers; the analytic arguments and numerical checks below do not change that status.

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1 Executive summary and nonduplication boundary

1.1 Corpus audit

The audit covered the living expositions, the consolidated frontier volume, the paper transcriptions, the glossary and Lean walkthrough, the currently visible frontier-draft families, and every frozen historical family listed by the repository archive manifest. The tree is actively edited, so the audit was performed at the document-family level: principal mathematical sources were read closely, draft reports were searched thematically and lexically, and auxiliary generated tables or frozen duplicates were checked through their parent documents and manifests. The principal interfaces are the current synthesis [10], the consolidated frontier volume [11], the inverse-frontier report [12], and the archive ledger [13].

The closest existing material is the repeated-integration dossier in the frontier volume. It studies how normalized repeated integrations of splines converge to up and gives exact convergence rates. Its object is a sequence of approximating functions. The present report instead fixes the limiting Fabius function and identifies its entire Volterra primitive tower. Searches for *antiderivative*, *indefinite integral*, and the corresponding monomial patterns returned no treatment of the formulas below in the current primary synthesis, consolidated frontier volume, or the new inverse, Newton, and spectral frontier reports.

All novelty labels in this report mean “not found in the recursively audited repository snapshot of 27 August 2026.” They are not claims that no equivalent identity has ever appeared elsewhere. Elementary consequences are included because they organize a useful calculus; the frontier claims concern the combined exact ladder, the all-complex-exponent convergence theorem, its Mellin and Bell consequences, and the inverse nonlinear reduction with its Thue–Morse realization.

1.2 Contribution map

2 Established platform and notation

2.1 The bounded Fabius function

We use the repository’s normalization. The bounded Fabius function $F : \mathbb{R} \rightarrow [0, 1]$ satisfies

$$F(x) = 0 \quad (x \leq 0), \quad F(x) = 1 \quad (x \geq 1), \quad (2.1)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad F(1-x) = 1 - F(x) \quad (x \in \mathbb{R}). \quad (2.2)$$

It is C^∞ , strictly increasing on $[0, 1]$, and flat at both endpoints. Let

$$X = \sum_{j \geq 1} 2^{-j} U_j, \quad U_j \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1]. \quad (2.3)$$

Then F is the distribution function of X . In particular, for every $z \in \mathbb{C}$, all complex moments $\mathbb{E}X^z$ exist after choosing the positive-real branch, because the law has infinite-order flatness at zero.

Write

$$d_n = \mathbb{E}X^n = n \int_0^1 t^{n-1} \text{up}(t) dt = n! 2^{\binom{n}{2}} F(2^{-n}), \quad n \in \mathbb{N}, \quad (2.4)$$

with $d_0 = 1$. The repository proves the triangular recurrence

$$(n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k, \quad (2.5)$$

Topic	Result	Status
Primitive ladder	n -fold zero-based primitive $2^{\binom{n}{2}} F(x/2^n)$ on $x \leq 1$, its signed global form, and its Volterra kernel.	Lean-formalized, more generally
Polynomial weights	Finite natural-monomial and arbitrary real-polynomial Volterra commutators.	Lean-formalized, more generally
Complex powers	Absolutely convergent exact series for every $\alpha \in \mathbb{C}$, with an exact remainder and compact-uniform convergence.	Frontier; not formalized
Mellin calculus	Entire Mellin transform, complete monotonicity on the real axis, dyadic interpolation, and logarithmic derivatives.	Frontier; not formalized
Arithmetic bridge	Complete moments as finite combinations of inverse-dyadic values, hence direct inputs for Bernoulli, Thue–Morse, and q -binomial formulae.	Frontier here; arithmetic inputs formalized
Rvachev transforms	Piecewise antiderivatives, global normalized primitives, and fractional absolute moments.	Frontier; not formalized
Inverse Fabius	Exact weighted reduction to $\int F^{\beta+1}$, critical logarithmic formula, convergence threshold, and order-statistic interpretation.	Frontier; not formalized
Nonlinear remainder	Explicit finite Thue–Morse spline formula for $\int F^r$, $r \in \mathbb{N}$.	Frontier; not formalized
Endpoint inverse	Weighted inverse-primitive equivalents, including the critical $\beta = -1$ law.	Frontier; informal consequence
Fractional primitive index	A continuous-index q -Volterra semigroup interpolating the integer ladder.	Frontier; conjectural

Table 1: Main contributions and status.

and the Bernoulli recurrence

$$d_n = \frac{n2^n}{4(2^n - 1)} d_{n-1} - \frac{1}{2^n - 1} \sum_{k=1}^{\lfloor n/2 \rfloor} \binom{n}{2k} 2^{n-2k} B_{2k} d_{n-2k}. \quad (2.6)$$

Thus any result expressed through the d_n is automatically connected to the corpus's Bernoulli and exact dyadic layers.

2.2 Rvachev's up-function and the signed extension

The even compactly supported bump is

$$\text{up}(x) = F(1 - |x|), \quad \text{supp up} = [-1, 1]. \quad (2.7)$$

Equivalently, up is the density of $Y = 2X - 1$. The signed global extension is

$$\mathcal{F}(x) = \sum_{m \geq 0} (-1)^{\text{wt}_2(m)} \text{up}(x - 2m - 1), \quad (2.8)$$

a locally finite sum satisfying

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (x \in \mathbb{R}). \quad (2.9)$$

It agrees with F on $(-\infty, 1]$ but is not the bounded distribution function to the right of 1. This distinction is essential when extending primitive formulas globally.

2.3 Inverse function

Let

$$J = F^{-1} : [0, 1] \longrightarrow [0, 1]. \quad (2.10)$$

Then $J(1 - y) = 1 - J(y)$ and J is smooth on $(0, 1)$, with infinite endpoint slope. The corpus establishes the leading lower-endpoint equivalent

$$J(y) \sim \frac{\sqrt{T}}{e\sqrt{\log 2}} \exp\left(-\sqrt{2(\log 2)T}\right), \quad T = \log \frac{1}{y} \longrightarrow \infty. \quad (2.11)$$

The sharper repository theory contains a periodic Gamma–zeta correction and Lambert– W reversion; (2.11) is the interface needed for the first endpoint laws below.

3 The exact primitive ladder

3.1 A self-similar Volterra chain

For $n \in \mathbb{N}_0$ define

$$A_0(x) = F(x), \quad A_n(x) = 2^{\binom{n}{2}} F(x/2^n) \quad (n \geq 1). \quad (3.1)$$

Theorem 3.1 (Primitive ladder). *For $x \leq 1$ and every $n \geq 1$,*

$$A'_n(x) = A_{n-1}(x), \quad A_n(0) = 0. \quad (3.2)$$

Consequently A_n is the n -fold zero-based primitive of F :

$$\boxed{2^{\binom{n}{2}} F(x/2^n) = \frac{1}{(n-1)!} \int_0^x (x-t)^{n-1} F(t) \, dt.} \quad (3.3)$$

For the signed extension $\mathcal{F}$, the same statements hold for every $x \in \mathbb{R}$.

Proof. On $(-\infty, 0)$ all the bounded candidates vanish identically, and flatness at zero supplies the same derivative identity there; the oriented Volterra integral also vanishes because F vanishes on $(-\infty, 0]$. For $0 \leq x \leq 1$ and $n \geq 1$, one has $0 \leq x/2^n \leq 1/2$, so (2.2) gives

$$A'_n(x) = 2^{\binom{n}{2}-n} F'(x/2^n) = 2^{\binom{n}{2}-n+1} F(x/2^{n-1}) = 2^{\binom{n-1}{2}} F(x/2^{n-1}) = A_{n-1}(x).$$

All $A_n(0)$ vanish. Iterating the fundamental theorem of calculus gives the Volterra kernel, with the negative-endpoint case interpreted as an oriented integral. For $\mathcal{F}$, equation (2.9) has no domain restriction, so the calculation is global. Equivalently, the bounded result on all of $(-\infty, 1]$ follows from the global one and $F = \mathcal{F}$ on that full initial half-line. $\square$

Remark 3.2 (Why the triangular exponent is inevitable). Let $(\mathcal{J}f)(x) = \int_0^x f(t) \, dt$ and $(Sf)(x) = f(x/2)$. Then

$$\mathcal{J}S = 2S\mathcal{J}, \quad \mathcal{J}F = SF. \quad (3.4)$$

Therefore

$$\mathcal{J}^n F = 2^{0+1+\dots+(n-1)} S^n F = 2^{\binom{n}{2}} S^n F. \quad (3.5)$$

The coefficients are thus a q -Weyl phenomenon with $q = 1/2$, not an accidental by-product of repeated differentiation. This is the operator-level bridge to the half-base q -binomial calculus in the corpus.

3.2 Repeated integration by parts

Theorem 3.3 (Finite weighted identity). *Let $g \in C^N([0, x])$, where $0 \leq x \leq 1$. Then*

$$\int_0^x g(t)F(t) dt = \sum_{k=0}^{N-1} (-1)^k g^{(k)}(x) A_{k+1}(x) + (-1)^N \int_0^x g^{(N)}(t) A_N(t) dt. \quad (3.6)$$

The identity is global with F, A_n replaced by $\mathcal{F}, 2^{\binom{n}{2}} \mathcal{F}(\cdot/2^n)$.

Proof. Use $A'_{k+1} = A_k$ and integrate by parts successively. Every lower boundary term vanishes because $A_k(0) = 0$. $\square$

4 Closed forms for nonnegative integer powers

4.1 The monomial formula

Theorem 4.1 (Polynomial-weight antiderivative). *For every $p \in \mathbb{N}_0$ and $x \leq 1$,*

$$\int x^p F(x) dx = \sum_{k=0}^p (-1)^k \frac{p!}{(p-k)!} 2^{\binom{k+1}{2}} x^{p-k} F(x/2^{k+1}) + C. \quad (4.1)$$

For the signed global extension $\mathcal{F}$, the same formula is valid for all $x \in \mathbb{R}$.

Proof. For $0 \leq x \leq 1$, apply [theorem 3.3](#) with $g(t) = t^p$ and $N = p + 1$. Since $g^{(p+1)} = 0$, the remainder vanishes. If $x \leq 0$, the oriented integral and every displayed scale term vanish because all their Fabius arguments are nonpositive. Adding an arbitrary constant converts the zero-based primitive to an indefinite antiderivative. $\square$

The first cases are

$$\int F(x) dx = F(x/2) + C, \quad (4.2)$$

$$\int xF(x) dx = xF(x/2) - 2F(x/4) + C, \quad (4.3)$$

$$\int x^2 F(x) dx = x^2 F(x/2) - 4xF(x/4) + 16F(x/8) + C, \quad (4.4)$$

$$\int x^3 F(x) dx = x^3 F(x/2) - 6x^2 F(x/4) + 48xF(x/8) - 384F(x/16) + C. \quad (4.5)$$

The cancellations are exact: differentiating the k th term produces one ordinary monomial derivative and one dilation derivative; the latter cancels the former contribution from the preceding scale.

Corollary 4.2 (Arbitrary polynomial weight). *For a polynomial P of degree m and $x \leq 1$,*

$$\int P(x)F(x) dx = \sum_{k=0}^m (-1)^k 2^{\binom{k+1}{2}} P^{(k)}(x) F(x/2^{k+1}) + C. \quad (4.6)$$

4.2 Continuation outside the unit interval

Formula (4.1) is a bounded- F identity on $(-\infty, 1]$. For $x > 1$, define the normalized primitive

$$I_p(x) = \int_0^x t^p F(t) dt. \quad (4.7)$$

Then

$$I_p(x) = I_p(1) + \frac{x^{p+1} - 1}{p+1}, \quad x \geq 1, \quad (4.8)$$

while $I_p(x) = 0$ for $x \leq 0$. Confusing this piecewise continuation with the global signed-extension formula is a normalization error: F is constant for $x \geq 1$, whereas $\mathcal{F}$ continues in alternating compact blocks.

Warning 4.3 (Formalization boundary beyond the natural finite core). The Lean theorems cover the bounded formulas through $x \leq 1$ and the signed formulas globally. The elementary bounded continuation (4.8) for $x > 1$ has not been packaged as a theorem in this antiderivative layer. Generic affine transport, basepoint shifts, and open-interval zero-tail Taylor jets are formalized, but they do not supply its explicit Fabius coefficients or its piecewise assembly. Nor have the later complex-power convergence theorem, Mellin continuation, fractional-index constructions, Rvachev primitive formulas, or inverse-function reductions been formalized. They remain frontier mathematics even where this report gives an analytic proof.

4.3 Complete moments and exact arithmetic

Putting $x = 1$ in (4.1) gives a finite inverse-dyadic formula.

Corollary 4.4 (Moment bridge). *For $p \in \mathbb{N}_0$,*

$$\int_0^1 x^p F(x) dx = \sum_{k=0}^p (-1)^k \frac{p!}{(p-k)!} 2^{\binom{k+1}{2}} F(2^{-k-1}) \quad (4.9)$$

$$= \frac{1}{p+1} \sum_{j=1}^{p+1} (-1)^{j-1} \binom{p+1}{j} d_j \quad (4.10)$$

$$= \boxed{\frac{1 - d_{p+1}}{p+1}}. \quad (4.11)$$

In particular, every integer-weight complete integral is rational.

Proof. The first equality is evaluation of the primitive. Use $2^{\binom{j}{2}} F(2^{-j}) = d_j/j!$ for the second. Finally, integration by parts against the probability measure dF gives

$$d_{p+1} = \int_0^1 x^{p+1} dF(x) = 1 - (p+1) \int_0^1 x^p F(x) dx.$$

□

Thus the antiderivative calculus is a direct consumer of every exact formula for $F(2^{-n})$ in the repository. For example, with $V_n = F(2^{-n})$,

$$V_0 = 1, \quad V_n = \frac{2^{-\binom{n}{2}}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}}}{(n-k+1)!} V_k, \quad (4.12)$$

so (4.9) is immediately executable in exact rational arithmetic. Alternatively one may insert the corpus's finite Thue–Morse formula or the translated q -binomial formula below. For $m, n \in \mathbb{N}$ and

arbitrary $Q \in \mathbb{C}$, its right-hand side is independent of Q (and equals the bounded value $F(m/2^n)$ whenever $m \leq 2^n$):

$$\begin{aligned} \mathcal{F}(m/2^n) &= \frac{1}{2^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \\ &\quad \times \sum_{r=0}^{m2^k-1} (-1)^{\text{wt}_2(r)} (r - m2^k + Q)^{n+k}, \end{aligned} \quad (4.13)$$

or the Bernoulli recurrence (2.6). No new summation theorem is required: the primitive formula provides the missing finite linear functional that transports those arithmetic representations into integration.

The first values are

$\int_0^1 x^p F(x) dx$	p	0	1	2	3	4	5
		$\frac{1}{2}$	$\frac{13}{36}$	$\frac{5}{18}$	$\frac{1207}{5400}$	$\frac{251}{1350}$	$\frac{22661}{142884}$

(4.14)

4.4 A scaled reflection identity

Because $F(1-x) = 1 - F(x)$, the same integrand has two different primitive representations. Equating them gives a family of functional identities.

Proposition 4.5 (Integrated reflection). *Let $I_j(x) = \int_0^x t^j F(t) dt$. For $p \in \mathbb{N}_0$ and $0 \leq x \leq 1$,*

$$\boxed{\sum_{j=0}^p (-1)^j \binom{p}{j} I_j(1-x) = I_p(x) - \frac{x^{p+1}}{p+1} + \frac{d_{p+1}}{p+1}}. \quad (4.15)$$

After inserting (4.1), this becomes a finite identity between values of F at the two dyadic scale towers $x/2^r$ and $(1-x)/2^r$.

Proof. Substitution $u = 1 - x$ gives

$$\int x^p F(1-x) dx = - \sum_{j=0}^p (-1)^j \binom{p}{j} I_j(1-x) + C.$$

The same integrand is $x^p(1 - F(x))$, with primitive $x^{p+1}/(p+1) - I_p(x)$. Determine the constant at $x = 1$ using (4.11). $\square$

5 Fractional, negative, and complex powers

5.1 Flatness estimate

The all-complex-exponent result rests on more than formal repeated integration.

Lemma 5.1 (Uniform Volterra flatness). *For every integer $s \geq 0$ there is a constant C_s such that, for all $n \geq 1$ and $0 \leq x \leq 1$,*

$$0 \leq A_n(x) \leq C_s \frac{s!}{(s+n)!} x^{s+n} = C_s \frac{x^{s+n}}{(s+1)_n}. \quad (5.1)$$

Proof. Endpoint flatness gives $F(t) \leq C_s t^s$ on $[0, 1]$. Insert this estimate into (3.3) and evaluate the beta integral:

$$A_n(x) \leq \frac{C_s}{(n-1)!} \int_0^x (x-t)^{n-1} t^s dt = C_s \frac{s!}{(s+n)!} x^{s+n}.$$

$\square$

5.2 The exact complex-exponent series

For $x > 0$ use the principal real logarithm and write $x^\alpha = \exp(\alpha \log x)$. Define

$$I_\alpha(x) = \int_0^x t^\alpha F(t) dt. \quad (5.2)$$

Unlike the corresponding integral with $1 - F$, this converges for every $\alpha \in \mathbb{C}$, because F is flat at zero.

Theorem 5.2 (All-complex-exponent antiderivative). *For every $\alpha \in \mathbb{C}$ and $0 < x \leq 1$,*

$$I_\alpha(x) = \sum_{k=0}^{\infty} (-1)^k \alpha^{\underline{k}} 2^{\binom{k+1}{2}} x^{\alpha-k} F(x/2^{k+1}). \quad (5.3)$$

The series is absolutely convergent and locally uniformly convergent in $(\alpha, x) \in \mathbb{C} \times (0, 1]$. It therefore defines an entire function of α . For $\alpha = p \in \mathbb{N}_0$ it terminates at $k = p$ and reduces to (4.1).

For every $N \geq 1$ the exact remainder after N terms is

$$R_N(\alpha, x) = (-1)^N \alpha^{\underline{N}} \int_0^x t^{\alpha-N} A_N(t) dt. \quad (5.4)$$

Proof. Apply [theorem 3.3](#) with $g(t) = t^\alpha$. This gives the first N terms and (5.4). Choose s so that $\Re \alpha + s + 1 > 0$. By [lemma 5.1](#),

$$|R_N(\alpha, x)| \leq \frac{C_s |\alpha^{\underline{N}}|}{(s+1)_N} \frac{x^{\Re \alpha + s + 1}}{\Re \alpha + s + 1}. \quad (5.5)$$

For nonintegral α , $\alpha^{\underline{N}} = (-1)^N \Gamma(N - \alpha) / \Gamma(-\alpha)$, so the quotient in (5.5) is $\mathcal{O}(N^{-\Re \alpha - s - 1})$. For nonnegative integer α it vanishes identically for large N . Uniform gamma-ratio estimates on compact α -sets, with the polynomial definition of the falling factorial used across its removable singularities, give compact-uniform convergence. The same bound applied termwise proves absolute convergence. Entire dependence under the integral also follows directly from dominated convergence, using flatness of arbitrarily high order. $\square$

Corollary 5.3 (One-step scale recurrence). *For every $\alpha \in \mathbb{C}$ and $0 < x \leq 1$,*

$$I_\alpha(x) = x^\alpha F(x/2) - \alpha 2^\alpha I_{\alpha-1}(x/2). \quad (5.6)$$

Here $2^\alpha = \exp(\alpha \log 2)$. Iterating (5.6) gives the finite expansion and exact remainder in [theorem 5.2](#); letting the depth tend to infinity gives (5.3).

Proof. Integrate once by parts with the primitive $F(t/2)$, and substitute $t = 2u$ in the remaining integral. Convergence at the lower endpoint holds for every α by flatness. $\square$

Remark 5.4 (A convergent integration-by-parts expansion). Ordinary repeated integration by parts often produces a divergent asymptotic series. Here the rapidly increasing dilation factor $2^{\binom{k+1}{2}}$ is exactly defeated by the much smaller value $F(x/2^{k+1})$. The Volterra beta-factor $(s+1)_{k+1}^{-1}$ exposes the mechanism. The result is a genuine convergent expansion for every complex exponent.

For $x > 1$ the bounded-function continuation is

$$I_\alpha(x) = I_\alpha(1) + \begin{cases} \frac{x^{\alpha+1} - 1}{\alpha + 1}, & \alpha \neq -1, \\ \log x, & \alpha = -1, \end{cases} \quad (5.7)$$

where the first quotient has its usual removable value at $\alpha = -1$. For $\mathcal{F}$ the series (5.3), with F replaced by $\mathcal{F}$, is valid for every $x > 0$ because the dilation equation is global.

5.3 Negative integer powers

When $\alpha = -m$, $m \in \mathbb{N}$, the two signs cancel because $-m^{\underline{k}} = (-1)^k m^{\bar{k}}$.

Corollary 5.5 (Positive-term negative-power series). *For $m \in \mathbb{N}$ and $0 < x \leq 1$,*

$$\int_0^x \frac{F(t)}{t^m} dt = \sum_{k=0}^{\infty} m^{\bar{k}} 2^{\binom{k+1}{2}} x^{-m-k} F(x/2^{k+1}). \quad (5.8)$$

All terms are nonnegative. In particular,

$$\int_0^x \frac{F(t)}{t} dt = \sum_{k=0}^{\infty} k! 2^{\binom{k+1}{2}} x^{-k-1} F(x/2^{k+1}). \quad (5.9)$$

This positivity gives certified lower bounds from partial sums. Combining (5.5) with an explicit, formally verified flatness constant would turn it into a fully explicit rational enclosure at dyadic x .

5.4 Logarithmic powers, Stirling numbers, and Bell polynomials

Since $I_\alpha(x)$ is entire,

$$\frac{\partial^r}{\partial \alpha^r} I_\alpha(x) = \int_0^x t^\alpha (\log t)^r F(t) dt. \quad (5.10)$$

Termwise differentiation in (5.3) gives

$$\begin{aligned} \int_0^x t^\alpha (\log t)^r F(t) dt &= \sum_{k \geq 0} (-1)^k 2^{\binom{k+1}{2}} F(x/2^{k+1}) x^{\alpha-k} \\ &\quad \times \sum_{\ell=0}^r \binom{r}{\ell} (\log x)^{r-\ell} \frac{d^\ell}{d\alpha^\ell} \alpha^{\underline{k}}. \end{aligned} \quad (5.11)$$

The derivative may be written without singularities through signed Stirling numbers $s(k, j)$:

$$\frac{d^\ell}{d\alpha^\ell} \alpha^{\underline{k}} = \sum_{j=\ell}^k s(k, j) j^\ell \alpha^{j-\ell}. \quad (5.12)$$

Away from the removable roots $0, 1, \dots, k-1$, it also has the Bell form

$$\frac{d^\ell}{d\alpha^\ell} \alpha^{\underline{k}} = \alpha^{\underline{k}} \mathcal{B}_\ell(h_1, \dots, h_\ell), \quad (5.13)$$

where $\mathcal{B}_\ell$ is the complete exponential Bell polynomial and

$$h_j = \psi^{(j-1)}(\alpha + 1) - \psi^{(j-1)}(\alpha - k + 1). \quad (5.14)$$

This supplies a direct antiderivative connection to the Bell-polynomial machinery used elsewhere in the corpus for saddle and inverse expansions.

At $\alpha = 0$, $\frac{d}{d\alpha} \alpha^{\underline{k}}|_{\alpha=0} = (-1)^{k-1} (k-1)!$ for $k \geq 1$, so

$$\int_0^x F(t) \log t dt = F(x/2) \log x - \sum_{k=1}^{\infty} (k-1)! 2^{\binom{k+1}{2}} x^{-k} F(x/2^{k+1}). \quad (5.15)$$

6 The Mellin transform as an entire dyadic interpolant

Define

$$\mathcal{M}_F(s) = \int_0^1 x^{s-1} F(x) dx. \quad (6.1)$$

Theorem 6.1 (Entire Mellin series). *The function $\mathcal{M}_F$ is entire and*

$$\mathcal{M}_F(s) = \sum_{k=0}^{\infty} (-1)^k s - 1^k 2^{\binom{k+1}{2}} F(2^{-k-1}). \quad (6.2)$$

For $\Re s > 0$,

$$\mathcal{M}_F(s) = \frac{1 - \mathbb{E}X^s}{s}. \quad (6.3)$$

Since $s \mapsto \mathbb{E}X^s$ is entire, this quotient continues to every $s \in \mathbb{C}$; its singularity at $s = 0$ is removable, with

$$\mathcal{M}_F(0) = -\mathbb{E} \log X. \quad (6.4)$$

Proof. Use [theorem 5.2](#) with $\alpha = s - 1$. For $\Re s > 0$, integration by parts against dF gives $\mathbb{E}X^s = 1 - s\mathcal{M}_F(s)$. Both sides are entire, so the identity continues to all s ; l'Hopital's rule gives (6.4). $\square$

Proposition 6.2 (Strict complete monotonicity and log-convexity). *For every real s and $r \in \mathbb{N}_0$,*

$$(-1)^r \mathcal{M}_F^{(r)}(s) = \int_0^1 x^{s-1} (-\log x)^r F(x) dx > 0. \quad (6.5)$$

Consequently $\mathcal{M}_F$ is strictly completely monotone on the entire real axis and strictly log-convex there.

Proof. Flatness justifies differentiation for every real s . Positivity is strict because the integrand is positive on a set of positive measure. Log-convexity follows from Cauchy–Schwarz applied to the measure $x^{s-1} F(x) dx$. $\square$

The change of variables $x = e^{-u}$ gives the bilateral-strength Laplace representation

$$\mathcal{M}_F(s) = \int_0^{\infty} e^{-su} F(e^{-u}) du, \quad (6.6)$$

which converges for every real s because $F(e^{-u})$ decays faster than every ordinary exponential. Thus the inverse-dyadic values in (6.2) interpolate an entire, positive, log-convex Mellin object rather than merely its integer samples.

7 The transforms $1 - F(x)$ and $F(1 - x)$

The bounded normalization satisfies $F(1 - x) = 1 - F(x)$ on all of $\mathbb{R}$. For $x > 0$ and $\alpha \neq -1$, an indefinite primitive is

$$\int x^\alpha (1 - F(x)) dx = \frac{x^{\alpha+1}}{\alpha+1} - I_\alpha(x) + C. \quad (7.1)$$

At $\alpha = -1$ it is

$$\int \frac{1 - F(x)}{x} dx = \log x - I_{-1}(x) + C. \quad (7.2)$$

The normalized integral from zero exists only for $\Re \alpha > -1$, because $1 - F(0) = 1$; this contrasts with I_α , which exists for every complex α .

For $p \in \mathbb{N}_0$, substitution $u = 1 - x$ gives the second finite form

$$\int x^p F(1 - x) dx = - \sum_{j=0}^p (-1)^j \binom{p}{j} I_j(1 - x) + C. \quad (7.3)$$

Equations (7.1) and (7.3) are equal up to the constant computed in [proposition 4.5](#).

For example,

$$\int F(1 - x) dx = x - F(x/2) + C, \quad (7.4)$$

$$\int xF(1 - x) dx = \frac{x^2}{2} - xF(x/2) + 2F(x/4) + C. \quad (7.5)$$

8 Rvachev up-function primitives

8.1 Piecewise polynomial weights

On $0 < x < 1$, $\text{up}(x) = 1 - F(x)$, so

$$\int x^p \text{up}(x) dx = \frac{x^{p+1}}{p+1} - I_p(x) + C. \quad (8.1)$$

On $-1 < x < 0$, $\text{up}(x) = F(1 + x)$ and, for integer p ,

$$\int x^p \text{up}(x) dx = \sum_{j=0}^p (-1)^{p-j} \binom{p}{j} I_j(1 + x) + C. \quad (8.2)$$

Theorem 8.1 (Global normalized polynomial primitive of up). *Let*

$$H_p(x) = \int_{-\infty}^x t^p \text{up}(t) dt. \quad (8.3)$$

Then

$$H_p(x) = \begin{cases} 0, & x \leq -1, \\ \sum_{j=0}^p (-1)^{p-j} \binom{p}{j} I_j(1 + x), & -1 \leq x \leq 0, \\ \frac{(-1)^p d_{p+1}}{p+1} + \frac{x^{p+1}}{p+1} - I_p(x), & 0 \leq x \leq 1, \\ \frac{1 + (-1)^p}{p+1} d_{p+1}, & x \geq 1. \end{cases} \quad (8.4)$$

Proof. The first two branches follow from (8.2). Evenness gives

$$\int_{-1}^0 t^p \text{up}(t) dt = (-1)^p \int_0^1 t^p \text{up}(t) dt = \frac{(-1)^p d_{p+1}}{p+1}.$$

Add the positive-side primitive (8.1). The final branch is the complete moment. $\square$

For $p = 0$ this gives the particularly simple distribution function of the up-density:

$$H_0(x) = \begin{cases} 0, & x \leq -1, \\ F((1 + x)/2), & -1 \leq x \leq 0, \\ \frac{1}{2} + x - F(x/2), & 0 \leq x \leq 1, \\ 1, & x \geq 1. \end{cases} \quad (8.5)$$

8.2 Fractional and negative powers

For $0 < r \leq 1$ define the positive-side complement primitive

$$\Phi_\alpha(r) = \begin{cases} \frac{r^{\alpha+1}}{\alpha+1} - I_\alpha(r), & \alpha \neq -1, \\ \log r - I_{-1}(r), & \alpha = -1. \end{cases} \quad (8.6)$$

Then $\Phi'_\alpha(r) = r^\alpha \text{up}(r)$ on $(0, 1)$. Hence for every complex α it is a local antiderivative on the positive half of the support. If on $-1 < x < 0$ one chooses the upper-half-plane branch $x^\alpha = e^{i\pi\alpha}(-x)^\alpha$, then

$$\int x^\alpha \text{up}(x) dx = \begin{cases} -e^{i\pi\alpha} \Phi_\alpha(-x) + C, & -1 < x < 0, \\ \Phi_\alpha(x) + C, & 0 < x < 1. \end{cases} \quad (8.7)$$

The lower-half-plane branch replaces $e^{i\pi\alpha}$ by $e^{-i\pi\alpha}$. Since I_α is given by (5.3), these are explicit convergent series on both open half-intervals, including every negative and fractional exponent.

A single primitive across the origin exists exactly when $\Re\alpha > -1$ for the branch-free weight $|x|^\alpha$. Put

$$T_\alpha = \Phi_\alpha(1) = \frac{\mathbb{E}X^{\alpha+1}}{\alpha+1}. \quad (8.8)$$

Then the global normalized primitive is

$$\int_{-1}^x |t|^\alpha \text{up}(t) dt = \begin{cases} 0, & x \leq -1, \\ T_\alpha - \Phi_\alpha(-x), & -1 \leq x \leq 0, \\ T_\alpha + \Phi_\alpha(x), & 0 \leq x \leq 1, \\ 2T_\alpha, & x \geq 1. \end{cases} \quad (8.9)$$

In particular,

$$\int_{-1}^1 |x|^\alpha \text{up}(x) dx = \frac{2\mathbb{E}X^{\alpha+1}}{\alpha+1}, \quad \Re\alpha > -1. \quad (8.10)$$

The right side supplies a meromorphic continuation to all $\alpha \in \mathbb{C}$, with a simple pole of residue 2 at $\alpha = -1$. The pole records the genuine nonintegrability of $|x|^{-1} \text{up}(x)$ at the central point, in contrast with the endpoint-flat integral I_{-1} .

For $m \in \mathbb{N}_0$,

$$\int_{-1}^1 x^{2m} \text{up}(x) dx = \frac{2d_{2m+1}}{2m+1}, \quad \int_{-1}^1 x^{2m+1} \text{up}(x) dx = 0. \quad (8.11)$$

Thus the antiderivative formula recovers the moment bridge between the Fabius distribution and the infinite sinc product $\widehat{\text{up}}(\xi) = \prod_{j \geq 1} \text{sinc}(2^{-j}\xi)$: the Taylor coefficients of the product are exactly the complete even moments in (8.11).

9 The signed global extension

Since (2.9) holds on all of $\mathbb{R}$, the cleanest version of the calculus belongs to $\mathcal{F}$.

Theorem 9.1 (Global polynomial and complex-power formulas). *For $p \in \mathbb{N}_0$ and all $x \in \mathbb{R}$,*

$$\int x^p \mathcal{F}(x) dx = \sum_{k=0}^p (-1)^k \frac{p!}{(p-k)!} 2^{\binom{k+1}{2}} x^{p-k} \mathcal{F}(x/2^{k+1}) + C. \quad (9.1)$$

For $x > 0$ and $\alpha \in \mathbb{C}$,

$$\int_0^x t^\alpha \mathcal{F}(t) dt = \sum_{k=0}^{\infty} (-1)^k \alpha^{\underline{k}} 2^{\binom{k+1}{2}} x^{\alpha-k} \mathcal{F}(x/2^{k+1}), \quad (9.2)$$

with absolute, locally uniform convergence.

The signed blocks in (2.8) therefore possess two complementary primitive descriptions: a finite scaled- $\mathcal{F}$ formula and a locally finite sum of shifted primitives of up. Equating them gives blockwise Thue–Morse identities that may be useful for formalizing the interaction between the global binary series and integration.

10 Integrals of the inverse Fabius function

10.1 The unweighted primitive

Theorem 10.1 (Exact inverse primitive). For $0 < y < 1$,

$$\boxed{\int J(y) dy = yJ(y) - F(J(y)/2) + C.} \quad (10.1)$$

The zero-based normalization is obtained with $C = 0$.

Proof. Set $a = J(y)$, so $y = F(a)$. Changing variables $t = F(u)$ gives

$$\int_0^y J(t) dt = \int_0^a uF'(u) du = ay - \int_0^a F(u) du = yJ(y) - F(J(y)/2).$$

The last step is (4.2). □

At $y = 1$, (10.1) gives $\int_0^1 J(y) dy = 1 - F(1/2) = 1/2$, as also follows from inverse symmetry.

10.2 Arbitrary complex weights

For $\Re\beta > -1$ define

$$K_\beta(y) = \int_0^y t^\beta J(t) dt. \quad (10.2)$$

Theorem 10.2 (Weighted inverse reduction). Let $0 < y \leq 1$ and put $a = J(y)$. If $\beta \neq -1$ and the integrals are interpreted on any common local interval avoiding zero, then

$$\boxed{\int y^\beta J(y) dy = \frac{y^{\beta+1} J(y)}{\beta+1} - \frac{1}{\beta+1} \int^{J(y)} F(x)^{\beta+1} dx + C.} \quad (10.3)$$

For $\Re\beta > -1$, the zero-based form is

$$\boxed{K_\beta(y) = \frac{y^{\beta+1} J(y)}{\beta+1} - \frac{1}{\beta+1} \int_0^{J(y)} F(x)^{\beta+1} dx.} \quad (10.4)$$

At the critical exponent $\beta = -1$,

$$\boxed{K_{-1}(y) = J(y) \log y - \int_0^{J(y)} \log F(x) dx.} \quad (10.5)$$

Proof. With $x = J(y)$,

$$K_\beta(y) = \int_0^a x F(x)^\beta F'(x) dx.$$

For $\beta \neq -1$, integrate $x d(F(x)^{\beta+1})/(\beta+1)$ by parts. At $\beta = -1$, integrate $x d \log F(x)$. The endpoint boundary $x \log F(x) \rightarrow 0$ follows from the repository's small-argument asymptotic (and already from sufficiently strong flatness bounds). $\square$

For $p \in \mathbb{N}_0$ this becomes

$$K_p(y) = \frac{y^{p+1} J(y)}{p+1} - \frac{1}{p+1} \int_0^{J(y)} F(x)^{p+1} dx. \quad (10.6)$$

The reduction is exact, but nonlinear: unlike the forward monomial primitive, it introduces a power of F . This identifies the precise missing object needed for a closed inverse calculus.

10.3 Endpoint integrability threshold

Proposition 10.3 (Sharp threshold). *The normalized integral $K_\beta(y)$ is absolutely convergent at zero exactly when*

$$\Re \beta \geq -1. \quad (10.7)$$

For $\Re \beta < -1$ it diverges, although the local antiderivative (10.3) remains valid away from zero.

Proof. Put $t = e^{-u}$. Up to a harmless phase,

$$\left| t^\beta J(t) dt \right| = e^{-(\Re \beta + 1)u} J(e^{-u}) du.$$

By (2.11), $J(e^{-u})$ decays like $\sqrt{u} e^{-\sqrt{2(\log 2)u}}$, which is subexponential in u but integrable. The assertion follows according as the ordinary exponential factor decays, is absent, or grows. $\square$

10.4 Complete inverse moments and order statistics

Let $M_r = \max(X_1, \dots, X_r)$ for iid copies of X . Since $\mathbb{P}(M_r \leq x) = F(x)^r$,

$$\int_0^1 F(x)^r dx = \mathbb{E}(1 - M_r). \quad (10.8)$$

Putting $y = 1$ in (10.6) gives

$$\boxed{\int_0^1 y^p J(y) dy = \frac{\mathbb{E} M_{p+1}}{p+1}.} \quad (10.9)$$

Thus inverse moments are normalized expected maxima of Fabius random variables. This probabilistic meaning is complementary to the algebraic inverse-dyadic and Lambert–Bell analysis in the repository's inverse-frontier report.

At the critical exponent,

$$\int_0^1 \frac{J(y)}{y} dy = - \int_0^1 \log F(x) dx, \quad (10.10)$$

an entropy-like constant attached to the distribution function rather than to its density.

11 A finite Thue–Morse formula for the nonlinear inverse remainder

The integral of F^r in (10.6) is not reduced by the linear primitive ladder. It nevertheless has an exact finite-level representation. Let

$$X_N = \sum_{j=1}^N 2^{-j} U_j \quad (11.1)$$

and let F_N be its distribution function. Inclusion–exclusion for unequal box splines gives

$$F_N(x) = \frac{2^{N(N+1)/2}}{N!} \sum_{m=0}^{2^N-1} (-1)^{\text{wt}_2(m)} (x - m/2^N)_+^N. \quad (11.2)$$

Moreover $F_N \rightarrow F$ uniformly.

For an integer $r \geq 1$, a cutoff $a \in [0, 1]$, and a tuple $\mathbf{m} = (m_1, \dots, m_r)$, put

$$b(\mathbf{m}) = \max_i \frac{m_i}{2^N}, \quad \delta_i = b(\mathbf{m}) - \frac{m_i}{2^N}. \quad (11.3)$$

Define

$$Q_{N,r}(a; \mathbf{m}) = \mathbf{1}_{\{a > b(\mathbf{m})\}} \sum_{k_1, \dots, k_r=0}^N \frac{\prod_{i=1}^r \binom{N}{k_i} \delta_i^{N-k_i}}{1 + k_1 + \dots + k_r} \times (a - b(\mathbf{m}))^{1+k_1+\dots+k_r}. \quad (11.4)$$

Theorem 11.1 (Finite Thue–Morse nonlinear integral). *For every $N, r \in \mathbb{N}$ and $a \in [0, 1]$,*

$$\int_0^a F_N(x)^r dx = \left(\frac{2^{N(N+1)/2}}{N!} \right)^r \sum_{m_1, \dots, m_r=0}^{2^N-1} (-1)^{\sum_i \text{wt}_2(m_i)} Q_{N,r}(a; \mathbf{m}). \quad (11.5)$$

$$\int_0^a F(x)^r dx = \lim_{N \rightarrow \infty} \int_0^a F_N(x)^r dx. \quad (11.6)$$

Consequently (10.6) has a completely explicit finite Thue–Morse spline limit for every $p \in \mathbb{N}_0$.

Proof. Expand the r th power of (11.2). For a fixed tuple, the product vanishes below $b = b(\mathbf{m})$. Above b , set $z = x - b$ and write

$$\prod_{i=1}^r (x - m_i/2^N)^N = \prod_{i=1}^r (z + \delta_i)^N.$$

The binomial theorem followed by termwise integration in z gives (11.4). Uniform convergence $F_N \rightarrow F$ and $0 \leq F_N, F \leq 1$ give (11.6) by dominated convergence. $\square$

The formula is exponentially expensive in both N and r . Its value is structural: it proves that the nonlinear remainder is still governed by the same finite Thue–Morse geometry, and it offers a starting point for orbit compression, Walsh transforms, and q -binomial acceleration.

12 Endpoint asymptotics for inverse primitives

Let

$$T = \log(1/y), \quad L = \log 2, \quad a = \sqrt{2L}, \quad C_0 = \frac{1}{e\sqrt{L}}. \quad (12.1)$$

Then (2.11) reads $J(y) \sim C_0 \sqrt{T} e^{-a\sqrt{T}}$.

Theorem 12.1 (Subcritical weighted inverse primitives). *Let $\rho = \beta + 1$ with $\Re \rho > 0$. As $y \downarrow 0$,*

$$\boxed{K_\beta(y) \sim \frac{y^\rho J(y)}{\rho}}. \quad (12.2)$$

For real $\rho > 0$, the leading model admits the refined endpoint denominator

$$K_\beta(y) = \frac{y^\rho J(y)}{\rho + \frac{a}{2\sqrt{T}} - \frac{1}{2T}} (1 + \mathcal{O}(T^{-1}) + \text{periodic-transfer terms}), \quad (12.3)$$

where a fully rigorous coefficient at the displayed relative order requires inserting the corpus's periodic next term for J .

Proof. After $t = e^{-u}$,

$$K_\beta(y) = \int_T^\infty e^{-\rho u} J(e^{-u}) du.$$

The factor $J(e^{-u})$ is slowly varying on additive u -scales compared with $e^{-\rho u}$. Direct one-sided Laplace estimation using (2.11) gives (12.2). Differentiating the leading exponent $\rho u + a\sqrt{u} - \frac{1}{2} \log u$ gives the denominator in (12.3); the repository's periodic correction must be retained before promoting the indicated structure to a complete second-term theorem. $\square$

At the critical exponent the ordinary exponential disappears and the scale changes.

Theorem 12.2 (Critical inverse primitive). *As $y \downarrow 0$,*

$$K_{-1}(y) \sim \frac{\sqrt{2}}{e \log 2} T \exp\left(-\sqrt{2(\log 2)T}\right) \quad (12.4)$$

$$\sim \sqrt{\frac{2T}{\log 2}} J(y). \quad (12.5)$$

Proof. Now

$$K_{-1}(y) = \int_T^\infty J(e^{-u}) du \sim C_0 \int_T^\infty \sqrt{u} e^{-a\sqrt{u}} du.$$

Set $v = \sqrt{u}$. The integral becomes $2C_0 \int_{\sqrt{T}}^\infty v^2 e^{-av} dv$, whose leading term is $(2C_0/a)T e^{-a\sqrt{T}}$. Substitute the constants. $\square$

For $\Re \beta < -1$, a primitive based at a fixed $y_0 > 0$ diverges at zero with leading magnitude $-y^{\beta+1} J(y)/(\beta+1)$. Thus $\beta = -1$ is a true transition: above it, the endpoint factor is $1/(\beta+1)$; at it, an extra $\sqrt{\log(1/y)}$ appears; below it, the same boundary term drives divergence.

13 Numerical experiments and exact verification

The accompanying script `experiments.py` is network-free and performs three logically separate checks.

1. It computes $V_n = F(2^{-n})$ exactly from (4.12) using rational arithmetic.
2. It represents $F(x/2^j)$ as a formal symbol F_j with derivative rule $F_j' = 2^{1-j} F_{j-1}$ and verifies the cancellation in (4.1) exactly for $0 \leq p \leq 12$.
3. It compares the series (5.3) at $x = 1$ with a scrambled Sobol simulation of the random series (2.3), using (6.3). This last item is only an independent numerical check.

The exact recurrence begins

$$V_0 = 1, \quad V_1 = \frac{1}{2}, \quad V_2 = \frac{5}{72}, \quad V_3 = \frac{1}{288}, \quad V_4 = \frac{143}{2073600}. \quad (13.1)$$

All formal derivative tests and the two moment evaluations (4.9)–(4.11) pass exactly.

α	series, 240 terms	Sobol mean	scramble spread
−1	0.7582400001750290	0.758239962896570	2.70×10^{-7}
−1/2	0.6073863107353257	0.607386310704139	2.51×10^{-8}
1/2	0.4209681733402818	0.420968175002922	4.38×10^{-10}
2.3	0.2591812728638399	0.259181273425061	5.26×10^{-11}

Table 2: Independent check of $I_\alpha(1)$. Each Sobol mean uses four scrambles of 2^{19} points and 28 random-series coordinates. The finite tail and QMC sampling error explain the displayed differences.

The slower apparent convergence at $\alpha = -1$ is expected: the term quotient is only polynomial after the superexponential dyadic factors have canceled. The analytic proof allows an arbitrarily strong polynomial majorant by choosing a higher flatness order s , but the practical constant then worsens. This is a natural target for Richardson acceleration based on the exact tail shape.

14 Further deductions

14.1 An exponential-weight generating primitive

Taking $g(x) = e^{zx}$ in the infinite version of [theorem 3.3](#) gives, for $0 < x \leq 1$,

$$\int_0^x e^{zt} F(t) dt = e^{zx} \sum_{k=0}^{\infty} (-z)^k 2^{\binom{k+1}{2}} F(x/2^{k+1}). \quad (14.1)$$

The series is entire in z . Expanding both sides recovers all polynomial primitives simultaneously. At $x = 1$ it provides a direct dyadic expansion for the integrated moment-generating function of F .

14.2 A uniqueness principle for the dyadic coefficient row

Suppose a family of coefficients c_k satisfies

$$\int P(x) F(x) dx = \sum_{k \geq 0} c_k P^{(k)}(x) F(x/2^{k+1}) + C$$

for all polynomials P , with only finitely many nonzero terms for each P . Testing successively on $1, x, x^2, \dots$ forces $c_k = (-1)^k 2^{\binom{k+1}{2}}$. Thus the triangular coefficient row in (4.6) is the unique local derivative functional built from the scale tower $F(x/2^{k+1})$.

14.3 A critical complement contrast

There is an instructive asymmetry at exponent -1 :

$$\int_0^x \frac{F(t)}{t} dt < \infty, \quad \int_0^x \frac{1 - F(t)}{t} dt = \infty. \quad (14.2)$$

The first has the positive series (5.9); the second has only the local primitive (7.2). After folding, the same obstruction is the pole of the fractional up-moment at $\alpha = -1$.

15 Conjectures and frontier directions

Conjecture 15.1 (Dyadic-tree algebra for nonlinear primitives). *For each integer $r \geq 2$, the functions*

$$Q_r(a) = \int_0^a F(x)^r dx$$

admit an absolutely convergent expansion indexed by finite rooted dyadic trees. The vertex weights should be the same inverse-dyadic moments d_n , while tree symmetries should produce partial Bell polynomials and half-base Gaussian binomial coefficients. The finite spline formula (11.5) is expected to collapse to this tree expansion after Walsh–Thue–Morse orbit compression.

Conjecture 15.2 (Periodic Lambert transfer for inverse primitives). *Let the sharp inverse expansion be written in the repository’s phase variable as*

$$J(y) = J_0(y) \left(1 + \frac{A_1(T, \theta)}{\sqrt{T}} + \frac{A_2(T, \theta)}{T} + \cdots \right),$$

where the A_j belong to the differential algebra generated by the periodic Gamma–zeta correction. Then, for every fixed $\Re \beta > -1$, $K_\beta(y)$ has an all-orders expansion in the same differential algebra, obtained by the formal resolvent

$$\left(\beta + 1 + \frac{d}{dT} \right)^{-1}.$$

At $\beta = -1$ the resolvent becomes singular and raises the polynomial degree in $\sqrt{T}$ by one, explaining (12.4).

Conjecture 15.3 (Sharp denominator law for polynomial primitives). *At a dyadic argument $x = a/2^n$, the reduced denominator of the finite primitive (4.1) has a predictable 2-adic valuation obtained by the minimum of the known valuations of $F(a/2^{n+k+1})$, except on explicitly classifiable cancellation congruence classes governed by Thue–Morse parity. The odd part should factor through the Mersenne products appearing in the half-base q -Pochhammer symbols.*

Conjecture 15.4 (Stieltjes structure of the negative-power row). *For each $m \in \mathbb{N}$ and fixed $x \in (0, 1]$, the positive terms in (5.8), after a natural factorial normalization, form a Stieltjes moment sequence. If true, diagonal Padé approximants would give alternating certified bounds and a substantially faster evaluation of $I_{-m}(x)$.*

Conjecture 15.5 (Continuous primitive index and dilation cocycle). *Let $A_\nu = \mathcal{J}^\nu F$ be the Riemann–Liouville fractional primitive, which already supplies the semigroup law $A_{\nu+\mu} = \mathcal{J}^\nu A_\mu$. There is a canonical transform $\mathcal{T}_\nu$, holomorphic in ν for $\Re \nu > 0$, and a scalar cocycle $c(\nu)$ such that*

$$A_\nu(x) = c(\nu) (\mathcal{T}_\nu F)(2^{-\nu}x), \quad c(n) = 2^{\binom{n}{2}}, \quad \mathcal{T}_n F = F \quad (n \in \mathbb{N}).$$

In Mellin coordinates, c should be expressible through a half-base q -gamma or Barnes-type function, while $\mathcal{T}_\nu$ should become the identity exactly at integer indices. The obstruction is that the integer formula uses the special vector F and does not arise from an operator identity on arbitrary functions.

15.1 Promising research program

The following projects are concrete and separable.

1. **Lean continuation.** The natural-order primitive ladder and the finite polynomial/monomial commutators are complete, including their stronger real-normed-target Volterra forms, affine covariance, and local basepoint/zero-tail Taylor-jet calculus. These generic results do not supply explicit Fabius/up coefficients or piecewise assembly. The next analytic boundary is the complex series, which requires an effective flatness lemma, a falling-factorial derivative library, and a uniform remainder theorem.

2. **Certified fractional evaluation.** Combine the exact rational recurrence (4.12) with an explicit formal flatness estimate to produce interval bounds for (5.3) at rational α .
3. **q -Richardson acceleration.** Determine the full asymptotic expansion of the positive tail in (5.8). The geometric scale suggests the $q = 1/4$ Lagrange filters already developed in the frontier corpus.
4. **Nonlinear Walsh compression.** Replace the 2^{Nr} tuple sum in (11.5) by orbit sums under coordinate permutation, bitwise XOR, and dyadic reflection; search for Gaussian-binomial collapse.
5. **Inverse periodic transfer.** Insert the established periodic Lambert–Bell expansion of J into the endpoint integrals and prove conjecture 15.2 with uniform phase derivatives.
6. **Fourier interpretation.** Express the even up-moments generated by (8.10) as fractional derivatives at the origin of the infinite sinc product, clarifying branch choices and possible Mellin–Barnes continuation.
7. **Arithmetic experiments.** Tabulate reduced denominators of (4.1) at general dyadic x and compare them with the formal 2-adic lower bounds and Mersenne factors in the current Lean corpus.

16 Lean crosswalk and remaining formalization frontier

The following are the exact current module and declaration names. The first module is independent of the Fabius function: it proves a reusable natural-order Volterra calculus for arbitrary real normed-space-valued inputs; the iteration/kernel bridge assumes completeness. This is strictly stronger than proving the report’s scalar identities one at a time.

Module	Declaration	Machine-checked content
FabiusFunction.Norm alizedVolterra	volterraPrimitive, iteratedPrimitive, normalizedVolterra	Literal iteration and normalized Cauchy kernels at arbitrary base points and oriented endpoints, including order zero.
FabiusFunction.Norm alizedVolterra	iteratedPrimitive_add, iteratedPrimitive_succ_hasSt rictDerivAt, iteratedPrimitiv e_eq_normalizedVolterra	Semigroup law, strict FTC ladder, and equality of literal iteration with the kernel formula for continuous inputs.
FabiusFunction.Norm alizedVolterra	normalizedVolterra_affine, no rmalizedVolterra_comp_affine	Forward affine covariance for arbitrary real normed targets and every real c , including zero and negative scales; the inverse form assumes $c \neq 0$.
FabiusFunction.Norm alizedVolterra	normalizedVolterra_basepoint _shift	Basepoint shift with interval integrability only from a to b and from b to x , no endpoint order, and every order including $n = 0$.
FabiusFunction.Norm alizedVolterra	normalizedVolterra_succ_eq_t aylor_of_eq_zero	Positive-order Taylor jet with integrability only from a to b and vanishing on the open unoriented interval between b and x ; neither endpoint order nor endpoint values are constrained.
FabiusFunction.Norm alizedVolterra	normalizedVolterra_kernel_pow	All-order kernel raising with the exact rising-factorial coefficient.
FabiusFunction.Norm alizedVolterra	normalizedVolterra_polynomial	Finite Hasse-derivative commutator for every real polynomial and every interval-integrable normed-space-valued input.
FabiusFunction.Norm alizedVolterra	normalizedVolterra_monomial	Natural-monomial specialization, with every natural Volterra order and all base/endpoint cases included.

Module	Declaration	Machine-checked content
FabiusFunction.GlobalExtension	fabiusReal_eq_extendedFabius_of_le_one	Equality of the bounded function and signed extension throughout $(-\infty, 1]$.
FabiusFunction.FabiusAntiderivatives	normalizedVolterra_extendedFabius	Complete signed primitive ladder for every natural order and every real endpoint.
FabiusFunction.FabiusAntiderivatives	normalizedVolterra_fabiusReal_of_le_one	Complete bounded primitive ladder on the proved initial half-line $x \leq 1$.
FabiusFunction.FabiusAntiderivatives	normalizedVolterra_pow_mul_extendedFabius	Finite natural-monomial formula for the signed extension, globally and at every natural Volterra order.
FabiusFunction.FabiusAntiderivatives	normalizedVolterra_pow_mul_fabiusReal_of_le_one	Finite natural-monomial formula for the bounded function when $x \leq 1$.

Thus the former placeholder declarations for the integer ladder and finite polynomial weights are retired. The generic affine and basepoint declarations do not calculate explicit Fabius/up coefficients or assemble report-specific piecewise branches. The remaining obligations are analytic, nonlinear, or explicit piecewise specializations:

- the complex-power series (5.3), including absolute and compact-local uniform convergence and a formal remainder bound;
- logarithmic weights and the entire Mellin theory derived from that series;
- fractional Volterra order and any semigroup interpolating the natural orders;
- the piecewise and fractional Rvachev formulas, including their Fourier sinc-product interpretations;
- every inverse-Fabius primitive, nonlinear-power reduction, Thue–Morse limit, and endpoint asymptotic in this report; and
- a named global piecewise primitive theorem for the bounded function to the right of 1 (distinct from the already formalized signed global ladder).

17 Conclusion

The Fabius differential equation contains an integration rule that is simpler than its differentiation rule: one integration merely halves the argument. Noncommutation of integration and halving accumulates the triangular factor $2^{\binom{n}{2}}$. Once this ladder is isolated, every polynomial-weighted antiderivative is finite, every complex-power antiderivative becomes a convergent series, and the Mellin transform becomes an entire dyadic interpolant. Reflection turns the same calculus into exact formulas for $1 - F(x)$, $F(1 - x)$, and up; the signed extension makes it global.

The inverse function has a different but equally clean structure. Its weighted primitives reduce exactly to nonlinear powers of F , revealing the specific obstacle to an elementary inverse calculus. Those nonlinear powers still possess a finite Thue–Morse spline limit, and the Lambert– W endpoint theory transfers to the primitives with a sharp critical transition at weight y^{-1} .

The main practical outcome is a compact toolbox: finite formulas for integer powers, convergent formulas for arbitrary complex powers, exact arithmetic at complete or dyadic endpoints, and a clear list of nonlinear and asymptotic problems remaining for future work.

A Proof details for compact-uniform convergence

Let $K \subset \mathbb{C}$ be compact and let $\sigma_- = \min_{\alpha \in K} \Re \alpha$. Choose an integer s with $s + \sigma_- + 1 > 1$. By [lemma 5.1](#), the absolute value of the k th term of (5.3) is bounded by

$$C_s x^{\sigma_- + s + 1} \frac{|\alpha^k|}{(s+1)_{k+1}}. \quad (\text{A.1})$$

On a compact set, the Pochhammer polynomial admits a uniform estimate

$$|\alpha^k| \leq C_K \Gamma(k - \sigma_- + c_K) \quad (\text{A.2})$$

for a fixed harmless c_K . Gamma-ratio asymptotics then give a summable majorant $C_{K,s} k^{-s-\sigma_- - 2 + c_K}$ after increasing s if necessary. The same calculation with $(s+1)_N$ rather than $(s+1)_{N+1}$ proves uniform vanishing of the remainder. Uniformity in x holds on every compact subinterval $[\varepsilon, 1]$; at $x = 0$ the primitive is identically zero but x^α itself has no compact-uniform meaning across arbitrary complex α .

B Corpus inventory by document family

The following ledger records the families used to establish the nonduplication boundary. The public tree is live; names are those visible on 27 August 2026.

Family	Role in the audit
Fabius_Function_and_Rvac hev_Up	Primary synthesis: definitions, probability, moments, exact inverse-dyadic and general dyadic values, Bernoulli recurrences, q -binomial formulas, global binary series, Fourier products, asymptotics, and Lean crosswalk.
non-formalized-research h-frontiers	Consolidated integration, repeated-integration, dyadic web, inverse, small-argument, Fourier-decay, and gap-register dossiers.
Glossary, non-elementarity, and Lean walkthrough	Normalization and notation checks; inverse and elementary-function boundaries; formalization status.
Paper transcriptions	Fabius, Arias de Reyna, arithmetic, lacunary-product, and related source material used by the repository.
Current draft families	Fabius_Arithmetic_Rays_Frontier_Report, Fabius_Dyadic_Self_Sampling_Frontier_Package, Fabius_Half_Integer_Spectral_Frontier_Report, Fabius_Inverse_Frontier_Report_Source_and_PDF, Fabius_Newton_Rvachev_Frontier_Report, the two Fabius_Rvachev_Frontier_Report families, Fabius_Rvachev_New_Frontiers, Fabius_Rvachev_Thue_Morse_Frontier_Results, Spectral_Arithmetic_Pascal_Rvachev_Hierarchy, Thue_Morse_Formula_Atlas, the dyadic-inverse-Barnes, new-results, results-bundle, spectral-endpoint, and rvachev_up_fourier_decay families. These set the newest nonduplication boundary for inverse, Bell, Mellin, q -Richardson, and sinc claims.

Family	Role in the audit
Archive manifest	Frozen early notes; standalone non-elementarity, small-argument, and inverse dyadic reports; six primary-exposition generations; two supplements; two q -calculus reports; two Lean reports; two glossary versions; and the frozen dyadic-asymptotic, alternative-representation, formula-to-asymptotic, q -connection, Thue–Morse convergence, gap-register, repeated-integration, and formula-atlas reports.
Auxiliary TeX fragments	Generated tables and display fragments in self-sampling and numerical bundles; checked through their parent reports and file inventories rather than treated as independent mathematical claims.

The audit found integration in the senses of moments, probability, quadrature, and construction by repeated integration. It did not find the scale-ladder antiderivative formulas, the exact all-complex-exponent series, or the weighted inverse reductions of this report.

C Reproducibility files

The accompanying archive contains:

- `Fabius_Monomial_Antiderivatives.tex`, this source;
- `Fabius_Monomial_Antiderivatives.pdf`, the rendered report;
- `experiments.py`, exact and numerical checks with detailed comments;
- `numerical_results.txt`, the full console report;
- `series_convergence.csv`, selected partial sums;
- `README.md` and `SHA256SUMS`.

Run

```
python experiments.py --output-dir .
```

to regenerate the numerical text and CSV files. The script requires Python 3, `mpmath`, `numpy`, and `scipy`; it makes no network requests.

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